ICE-4107: Microwave Engineering

Lecture 5

Electromagnetic Plane Waves

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ε= Epsilon
μ= Miu

Plane Waves in a Lossless Medium

- In previous course we have studied **Plane Waves in a Lossless Medium** and **Plane Waves in a Lossy Medium.**
- In a lossless medium, ε and μ are real numbers, and so k is real. A basic plane wave solution to the above wave equations can be found by considering an electric field with only an xˆ component and uniform (no variation) in the x and y directions. Then, ∂/∂x = ∂/∂y = 0, and the Helmholtz equation of (1.42 discussed in the previous slide) reduces to

$$\frac{\partial^2 E_x}{\partial z^2} + k^2 E_x = 0. \qquad (1.44)$$

Plane Waves in a Lossless Medium

The two independent solutions to this equation are easily seen, by substitution, to be of the form

$$E_x(z) = E^+ e^{-jkz} + E^- e^{jkz}, \qquad (1.45)$$

where E^+ and E^- are arbitrary amplitude constants. The above solution is for the time harmonic case at frequency ω. In the time domain, this result is written as

$$\mathcal{E}_x(z, t) = E^+ \cos(\omega t - kz) + E^- \cos(\omega t + kz), \qquad (1.46)$$

where we have assumed that E^+ and E^- are real constants

https://www.physicsforums.com/threads/conduction-and-lossless-materials.559039/

The *phase velocity* is the velocity at which a fixed phase point on the wave travels, and it is given by

$$v_p = \frac{dz}{dt} = \frac{d}{dt}\left(\frac{\omega t - \text{constant}}{k}\right) = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \tag{1.47}$$

The *wavelength*, λ, is defined as the distance between two successive maxima (or minima, or any other reference points) on the wave at a fixed instant of time. Thus,

$$(\omega t - kz) - [\omega t - k(z + \lambda)] = 2\pi,$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi v_p}{\omega} = \frac{v_p}{f}. \tag{1.48}$$

EXAMPLE 1.1 BASIC PLANE WAVE PARAMETERS

A plane wave propagating in a lossless dielectric medium has an electric field given as $\mathcal{E}_x = E_0 \cos(\omega t - \beta z)$ with a frequency of 5.0 GHz and a wavelength in the material of 3.0 cm. Determine the propagation constant, the phase velocity, the relative permittivity of the medium, and the wave impedance.

Solution

From (1.48) the propagation constant is $k = \dfrac{2\pi}{\lambda} = \dfrac{2\pi}{0.03} = 209.4 \text{ m}^{-1}$, and from (1.47) the phase velocity is

$$v_p = \frac{\omega}{k} = \frac{2\pi f}{k} = \lambda f = (0.03)\,(5 \times 10^9) = 1.5 \times 10^8 \text{ m/sec.}$$

This is slower than the speed of light by a factor of 2.0. The relative permittivity of the medium can be found from (1.47) as

$$\epsilon_r = \left(\frac{c}{v_p}\right)^2 = \left(\frac{3.0 \times 10^8}{1.5 \times 10^8}\right)^2 = 4.0$$

The wave impedance is

$$\eta = \eta_0/\sqrt{\epsilon_r} = \frac{377}{\sqrt{4.0}} = 188.5\ \Omega$$

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Plane Waves in a General Lossy Medium

If the medium is conductive, with a conductivity σ, Maxwell's curl equations can be written

$$\nabla \times \bar{E} = -j\omega\mu\bar{H}, \tag{1.50a}$$

$$\nabla \times \bar{H} = j\omega\epsilon\bar{E} + \sigma\bar{E}. \tag{1.50b}$$

The resulting wave equation for $\bar{E}$ then becomes

$$\nabla^2\bar{E} + \omega^2\mu\epsilon\left(1 - j\frac{\sigma}{\omega\epsilon}\right)\bar{E} = 0, \tag{1.51}$$

We then define a complex propagation constant for the medium as

$$\gamma = \alpha + j\beta = j\omega\sqrt{\mu\epsilon}\sqrt{1 - j\frac{\sigma}{\omega\epsilon}} \tag{1.52}$$

where α is the attenuation constant and β is the phase constant.

Plane Waves in a General Lossy Medium

If we again assume an electric field with only an xˆ component and uniform in x and y, the wave equation of (1.51) reduces to

$$\frac{\partial^2 E_x}{\partial z^2} - \gamma^2 E_x = 0, \tag{1.53}$$

which has solutions

$$E_x(z) = E^+ e^{-\gamma z} + E^- e^{\gamma z}. \tag{1.54}$$

The positive traveling wave then has a propagation factor of the form

$$e^{-\gamma z} = e^{-\alpha z} e^{-j\beta z},$$

which in the time domain is of the form

$$e^{-\alpha z} \cos(\omega t - \beta z).$$

Conductivity

σ = Sigma

- Conductivity is a measure of how easily electric current can flow through a given material. That is, for a given Electric Field in a material, a higher conductivity material will produce more current flow than a low conductivity material.
- Conductivity is the measure of the ease at which an electric charge or heat can pass through a material.
- Conductivity represents power loss within a material. A material is known as "lossless" if the conductivity is zero ($\sigma = 0$). Materials like air and vacuum (space) have no conductivity.
- When the conductivity is not zero, then an Electric Field flowing through the material will induce an Electric Current Density (**J**).

Conductivity

- The relationship between **E**, **J**, and is known as Ohm's Law and is given in Equation:

$$\mathbf{J} = \sigma \mathbf{E}$$

- In good conductors E decreases very fast because electrons can move freely in good conductor. Any E developed inside a good conductor will immediately cause electrons to move in direction that will neutralize the E field. So, the E will attenuate fast. For perfect conductor, no E field can even develop inside and therefore the skin depth is zero and all current are on the surface.

Conductivity

- As conductance goes up, attenuation constant goes up so EM wave disappeared quickly when penetrating into a good conductor. Also, the velocity of the EM propagation slow down quite a bit as conductance goes up.
- Some materials, such as copper or steel (or metals in general) have very high conductivities, and the conductivity can often be approximated as infinite. This means the material has a resistance of zero.
- For these materials, we can see from the Equation that the Electric Field must zero within these materials. If it was not zero, then from Equation the current density would be infinite, which is not the case.

Conductivity

- Hence, while we can have current flow through metals or high conductivity materials, the Electric Field inside these materials must be zero.
- In textbooks, these materials are often referred to as PEC (perfect electric conductors), so that the conduction loss can be ignored.
- Consider now materials with a conductivity greater than zero, but also non-metallic materials so that the conductivity is not extremely high.
- In this case, we may have electric current flowing in a material, along with an associated Electric Field.

σ = Sigma

Conductivity

- When electric current flows through a material, some of the energy is converted to heat (the energy is then lost from the Electromagnetic Wave or current).
- Materials with medium-range conductivities are known as lossy materials. An example of a lossy material with a mid-range value for conductivity is Carbon.
- Generally, we can divide materials into 3 regions by their conductivity

$\sigma = 0$	Lossless Materials (air, vacuum)
$\sigma > 0$	Lossy Materials (Salt Water, Silicon)
$\sigma >> 0$ $(\sigma \approx \infty)$	Conductors (Metals, Copper, Silver, Etc)

http://maxwells-equations.com/materials/conductivity.php

Medium Properties

- Medium is homogeneous: ε and μ are constant.
- Medium is linear: ε and μ will not change with the field
- Medium is isotropic: ε and μ do not depend on the direction of the field
- Medium is source free: No conduction current ($\sigma = 0$)

η = Eta
β = Beta

Uniform Plane-Wave Reflection

Normal-incidence reflection. The simplest reflection problem is that of a uniform plane wave normally incident on a plane boundary between two dielectric media with no surface-charge density and surface-current density. This situation is shown in Fig. 2-3-3.

In medium 1 the fields are the sum of an incident wave plus a reflected wave, respectively.

$$E_x^{(1)} = E_0(e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z}) \qquad (2\text{-}3\text{-}11)$$

$$H_y^{(1)} = \frac{E_0}{\eta_1}(e^{-j\beta_1 z} - \Gamma e^{j\beta_1 z}) \qquad (2\text{-}3\text{-}12)$$

where $\beta_1 = \omega\sqrt{\mu_1\epsilon_1}$

$\eta_1 = \sqrt{\dfrac{\mu_1}{\epsilon_1}} = \dfrac{\eta_0}{\sqrt{\epsilon_{r1}}}$ = intrinsic wave impedance of medium 1

Γ = reflection coefficient

Uniform Plane-Wave Reflection

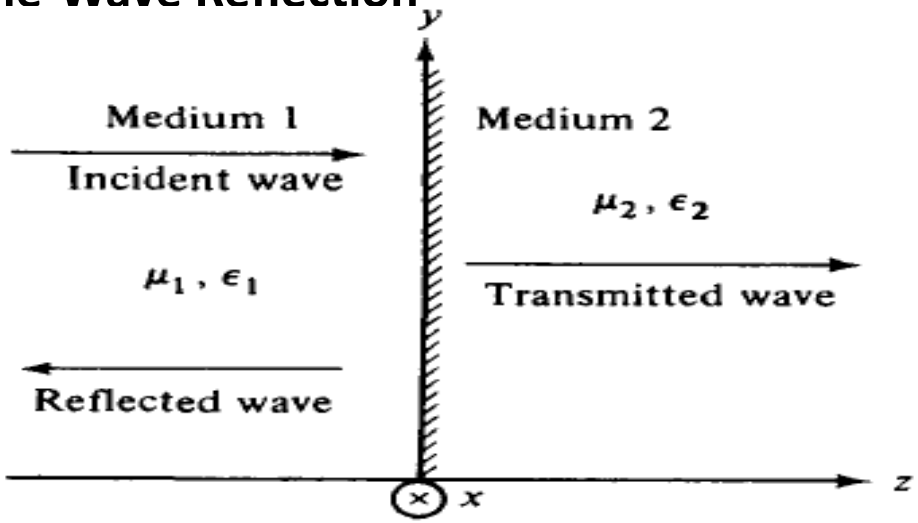


Figure 2-3-3 Uniform plane-wave reflection.

In medium 2 there are transmitted waves:

$$E_x^{(2)} = E_0 T e^{-j\beta_2 z} \tag{2-3-13}$$

$$H_y^{(2)} = \frac{E_0}{\eta_2} T e^{-j\beta_2 z} \tag{2-3-14}$$

where $\beta_2 = \omega\sqrt{\mu_2\epsilon_2}$

$\eta_2 = \sqrt{\dfrac{\mu_2}{\epsilon_2}} = \dfrac{\eta_0}{\sqrt{\epsilon_{r2}}}$ = intrinsic wave impedance of medium 2

T = transmission coefficient

Uniform Plane-Wave Reflection

For continuity of wave impedance at the boundary, the wave impedance is

$$Z_z = \frac{E_x^{(1)}}{H_y^{(1)}}\bigg|_{z=0} = \eta_1 \frac{1+\Gamma}{1-\Gamma} = \frac{E_x^{(2)}}{H_y^{(2)}}\bigg|_{z=0} = \eta_2 \tag{2-3-15}$$

Hence the reflection coefficient is given by

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \tag{2-3-16}$$

From the boundary condition the tangential components of electric field intensity are continuous across the interface. Then

$$E_x^{(1)}\bigg|_{z=0} = E_0(1+\Gamma) = E_x^{(2)}\bigg|_{z=0} = E_0 T \tag{2-3-17}$$

Hence the transmission coefficient is expressed as

$$T = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1} \tag{2-3-18}$$

If medium 1 is lossless dielectric (that is, $\sigma = 0$), the standing-wave ratio is defined as

$$\text{SWR} = \rho = \frac{|E_{\max}^{(1)}|}{|E_{\min}^{(1)}|} = \frac{1+\Gamma}{1-\Gamma} \tag{2-3-19}$$

Uniform Plane-Wave Reflection

The power density transmitted across the boundary is

$$p_{tr} = \frac{1}{2}(\mathbf{E} \times \mathbf{H}^*)\Big|_{z=0} \cdot \mathbf{u}_z = \frac{E_0^2}{2\eta_1}(1 - \Gamma^2) \qquad (2\text{-}3\text{-}20)$$

Then

$$p_{tr} = p_{inc}(1 - \Gamma^2) \qquad (2\text{-}3\text{-}21)$$

where p_{inc} is the incident power density.

The incident power density minus the transmitted power density would yield the reflected power density as

$$p_{ref} = p_{inc}\Gamma^2 \qquad (2\text{-}3\text{-}22)$$

Oblique-incidence reflection: E is in the Plane of Incidence.

- The plane of incidence is defined by the direction of propagation and the line normal to the boundary.
- The linearly polarized uniform plane waves with E lying in and **H** normal to the plane of incidence are impinging obliquely on a boundary between two lossless dielectric materials as shown in Fig. 2-3-4.

Whenever a wave is incident obliquely on the boundary surface between two media, the polarization of the wave is vertical or horizontal if the electric field is normal to or parallel to the boundary surface. The terms *horizontal* and *vertical polarization* refer to the phenomenon of waves from horizontal and vertical antennas, respectively, producing the corresponding orientations of wave polarization when the waves strike the surface of the earth.

Oblique-incidence reflection: E is in the Plane of Incidence.

For guided waves in waveguides, the terms *transverse electric* (TE) and *transverse magnetic* (TM) are used to designate the fact that either the electric or the magnetic field is normal to the direction of propagation. The polarization of a wave is an extremely useful concept for computing electromagnetic power flow. For example, a Poynting vector indicates that the power flow density is the cross product of an electric and a magnetic field with the specific direction determined by the polarizations of the two fields.

As Fig. 2-3-4 shows, for a lossless dielectric medium, the phase constants (or propagation constants) of the two media in the x direction on the interface are equal as required by the continuity of tangential E and H on the boundary. Thus

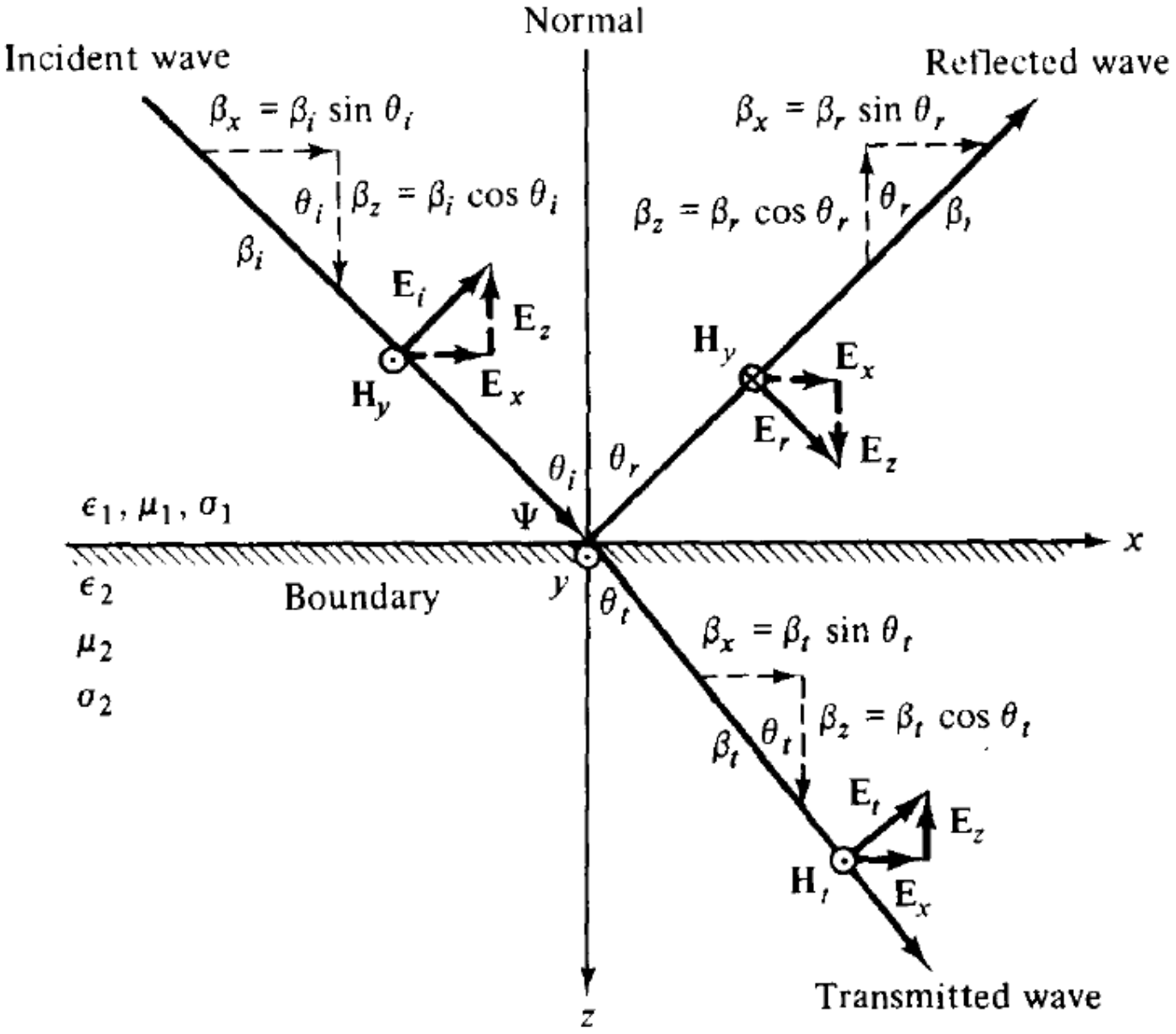


Figure 2-3-4 Reflection and transmission of oblique incidence.

$$\beta_i \sin \theta_i = \beta_r \sin \theta_r \tag{2-3-23a}$$

$$\beta_i \sin \theta_i = \beta_t \sin \theta_t \tag{2-3-23b}$$

From Eq. (2-3-23a), since $\beta_i = \beta_r = \beta_1$, the angle of reflection is equal to the angle of incidence. This is

$$\theta_1 = \theta_r \tag{2-3-24}$$

From Eq. (2-3-23b)

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{\beta_i}{\beta_t} = \frac{v_2}{v_1} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} \tag{2-3-25}$$

where v represents the phase velocity. This is well known as Snell's law. In general, all low-loss dielectrics have equal permeability—that is, $\mu_1 = \mu_2 = \mu_0$. If medium 2 is free space and medium 1 is a nonmagnetic dielectric, the right-hand side of Eq. (2-3-25) becomes $\sqrt{\epsilon_r}$, which is called the *index of refraction* of the dielectric.

The components of electric intensity **E** are

$$E_x = E_0 \cos \theta_i e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \tag{2-3-26}$$

$$E_y = 0 \tag{2-3-27}$$

$$E_z = -E_0 \sin \theta_i e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \tag{2-3-28}$$

The components of magnetic intensity **H** are

$$H_x = 0 \tag{2-3-29}$$

$$H_y = \frac{E_0}{\eta_1} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)} \tag{2-3-30}$$

$$H_z = 0 \tag{2-3-31}$$

The wave impedance in the z direction is given by

$$Z_z = \frac{E_x}{H_y} = \eta \cos \theta \tag{2-3-32}$$

It should be noted that the subscripts of η and θ have dropped because the wave impedances of the two regions in the z direction are the same.

The wave impedance can be expressed in terms of the reflection coefficient of the normal components at the boundary $z = 0$. In medium 1

$$E_x^{(1)} = E_0 \cos\theta_i \left[e^{-j\beta_1 z\cos\theta_i} + \Gamma e^{j\beta_1 z\cos\theta_r}\right] \tag{2-3-33}$$

$$H_y^{(1)} = \frac{E_0}{\eta_1}\left[e^{-j\beta_1 z\cos\theta_i} - \Gamma e^{j\beta_1 z\cos\theta_r}\right] \tag{2-3-34}$$

$$Z_z = \left.\frac{E_x^{(1)}}{H_y^{(1)}}\right|_{z=0} = \eta_1 \cos\theta_i \frac{1+\Gamma}{1-\Gamma} \tag{2-3-35}$$

The impedance must be equal to the z-directed wave impedance in region 2 at the boundary. Substitution of $Z_z = \eta_2 \cos\theta_t$ in Eq. (2-3-35) yields

$$\Gamma = \frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i} \tag{2-3-36}$$

Then the transmission coefficient is given by

$$T = \frac{2\eta_2\cos\theta_t}{\eta_2\cos\theta_t + \eta_1\cos\theta_i} \tag{2-3-37}$$

The preceding two equations are known as Fresnel's formulas for **E** in the plane of incidence.

- **Oblique-incidence reflection:** *H is in the Plane of Incidence*.

If **H** is in the plane of incidence, the components of **H** are:

$$H_x = H_0 \cos\theta_i e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)} \quad (2\text{-}3\text{-}38)$$

$$H_y = 0 \quad (2\text{-}3\text{-}39)$$

$$H_z = -H_0 \sin\theta_i e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)} \quad (2\text{-}3\text{-}40)$$

The components of electric intensity **E** normal to the plane of incidence are

$$E_x = 0 \quad (2\text{-}3\text{-}41)$$

$$E_y = -\eta_1 H_0 e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)} \quad (2\text{-}3\text{-}42)$$

$$E_z = 0 \quad (2\text{-}3\text{-}43)$$

The wave impedance in the z direction is given by

$$Z_z = -\frac{E_y}{H_x} = \frac{\eta}{\cos\theta} = \eta \sec\theta \quad (2\text{-}3\text{-}44)$$

- **Oblique-incidence reflection:** *H is in the Plane of Incidence*.

It should be noted that the subscripts of η and θ have been dropped for the same reason stated previously.

Fresnel's formulas for **H** in the plane of incidence are

$$\Gamma = \frac{\eta_2 \sec\theta_t - \eta_1 \sec\theta_i}{\eta_2 \sec\theta_t + \eta_1 \sec\theta_i} \tag{2-3-45}$$

$$T = \frac{2\eta_2 \sec\theta_t}{\eta_2 \sec\theta_t + \eta_1 \sec\theta_i} \tag{2-3-46}$$

Thank You